MATH2241 - Introduction to mathematical analysis

(1st semester 2026-2027)

Tutorial 1 Solution

1. Suppose that a, b, ϵ are in an ordered field F and $a < b + \epsilon$ for any $\epsilon > 0$. Prove that $a \leq b$.

   Solution.
   Suppose $b < a$, then take $\epsilon = a - b > 0$. We have

   $$a < b + (a - b)$$
   $$a < a$$

   Contradiction!

2. For each of the following sets, find the supremum and infimum (without proof) if they exist. State also whether the supremum and infimum are the maximum and minimum respectively.

   (a) $S_1 = \left\{x \in \mathbb{R} : x \neq 0, \frac{1}{x} > 2\right\}$

   (b) $S_2 = \left\{\frac{1}{k} + \frac{2}{m} + \frac{3}{n} : k, m, n \in \mathbb{N}\right\}$

   (c) $S_3 = \{3^x : x \in \mathbb{Z}\}$

   Solution.

   (a) $\sup S_1 = \frac{1}{2} \neq \max S_1$, $\inf S_1 = 0 \neq \min S_1$.

   (b) $\sup S_2 = 6 = \max S_2$, $\inf S_2 = 0 \neq \min S_2$.

   (c) $\sup S_3 = \infty \neq \max S_3$, $\inf S_3 = 0 \neq \min S_3$.

3. Let A be a nonempty bounded subset of $\mathbb{R}$. What can we say about A if $\inf A = \sup A$?

   Solution.
   If A contains two numbers a, b and $a < b$, then $\inf A \leq a < b \leq \sup A$, contradiction. A is a single element set.

4. Let A and B be nonempty bounded subsets of $\mathbb{R}$. Prove that if $A \subseteq B$, then

   $$\inf B \leq \inf A \leq \sup A \leq \sup B$$

   Solution.
   If $\inf B > \inf A$, let $r = \inf B - \inf A > 0$. There is $w \in A$ such that $w - \inf A < r \implies w < \inf B \implies w \notin B$, contradiction. The last inequality is similar, the middle one is by definition.

5. Let A and B be nonempty bounded subsets of $\mathbb{R}$. Define

$$A + B = \{a + b : a \in A \text{ and } b \in B\}$$
$$AB = \{a \times b : a \in A \text{ and } b \in B\}$$

   (a) Is that $\sup(A + B) = \sup A + \sup B$?

   (b) Is that $\sup(AB) = (\sup A) \times (\sup B)$?

   Solution.

   (a) $\forall x \in A + B$, $x = a + b \leq \sup A + \sup B$ for some $a \in A$ and $b \in B$. Now $\sup A + \sup B$ is an upper bound of $A + B$. If $\sup(A + B) < \sup A + \sup B$, take $r = \sup A + \sup B - \sup(A + B) > 0$. There are $a' \in A$, $b' \in B$ such that $\sup A - a' < r/2$ and $\sup B - b' < r/2$, then $\sup(A + B) < a' + b' \in A + B$, contradiction.

   (b) Let $A = \{1, 2\}$, $B = \{-2, -1\}$, then $AB = \{-4, -2, -1\}$. It's not true.

6. Let A be a nonempty subset of $\mathbb{R}$ and let w be an upper bound of A. Suppose that for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $w - \frac{1}{n} \leq a_n$. Prove that $w = \sup A$.

   Solution.
   If $r = w - \sup A > 0$, there is $n \in \mathbb{N}$ such that $r > \frac{1}{n} > 0$. Then there is $a_n \in A$ such that

$$\begin{aligned} a_n &\geq w - \frac{1}{n} \\ &\geq w - r \\ &= \sup A \end{aligned}$$

7. (a) Give an example of a set of irrational numbers whose supremum is rational.

   (b) Give an example of a set of rational numbers whose supremum is irrational.

   Solution.

   (a) $S_i = \left\{\frac{3}{\pi}, \frac{3.1}{\pi}, \frac{3.14}{\pi}, \frac{3.141}{\pi}, \dots\right\}$, $\sup S_i = 1$

   (b) $S_r = \{3, 3.1, 3.14, \dots\}$, $\sup S_r = \pi$

8. Show that

   (a) the root $\sqrt[m]{n}$ for $m, n \in \mathbb{N}$ is either an integer or an irrational number.

   (b) $\sqrt{n} + \sqrt{n+1}$ is an irrational number for every $n \in \mathbb{N}$.

   (c) $\sqrt{n + \sqrt{n}}$ is an irrational number for every $n \in \mathbb{N}$.

   Solution.

(a) Suppose $\sqrt[m]{n} \in \mathbb{Q}$ is not an integer. Write $\sqrt[m]{n} = \frac{p}{q}$, where $p, q \in \mathbb{Z}$, $\gcd(p, q) = 1$, and $q > 1$.

$$n = \frac{p^m}{q^m} \implies nq^m = p^m$$

Since $q > 1$, take a prime k such that $k \mid q$.

$$\begin{aligned} k \mid q &\implies k \mid q^m \implies k \mid nq^m \\ &\implies k \mid p^m \\ &\implies k \mid p \end{aligned}$$

Thus k divides both p and q, contradicting $\gcd(p, q) = 1$. Therefore, $\sqrt[m]{n}$ must be irrational if it is not an integer.

(b) Suppose $\sqrt{n} + \sqrt{n+1} = r \in \mathbb{Q}$. Taking the reciprocal gives:

$$\frac{1}{r} = \frac{1}{\sqrt{n+1} + \sqrt{n}} = \sqrt{n+1} - \sqrt{n}$$

Subtracting this from r:

$$r - \frac{1}{r} = 2\sqrt{n} \implies \sqrt{n} = \frac{r^2 - 1}{2r} \in \mathbb{Q}$$

By part (a), since $\sqrt{n} \in \mathbb{Q}$, n must be a perfect square ($n = k^2$ for $k \in \mathbb{N}$).

Similarly, adding the two equations:

$$r + \frac{1}{r} = 2\sqrt{n+1} \implies \sqrt{n+1} = \frac{r^2 + 1}{2r} \in \mathbb{Q}$$

So $n + 1$ must also be a perfect square ($n + 1 = m^2$ for $m \in \mathbb{N}$).

Then $m^2 - k^2 = 1 \implies (m - k)(m + k) = 1$, which has no integer solutions for $n \in \mathbb{N}$. Contradiction. Thus, $\sqrt{n} + \sqrt{n+1}$ is irrational.

(c) Suppose $\sqrt{n + \sqrt{n}} = r \in \mathbb{Q}$. Squaring both sides yields:

$$n + \sqrt{n} = r^2 \implies \sqrt{n} = r^2 - n$$

Since $r \in \mathbb{Q}$ and $n \in \mathbb{N}$, the right-hand side $r^2 - n$ is rational, so $\sqrt{n} \in \mathbb{Q}$.

By part (a), because $\sqrt{n} \in \mathbb{Q}$, n must be a perfect square. Thus, $n = k^2$ for some $k \in \mathbb{N}$.

Substituting $n = k^2$ back into the original expression:

$$\sqrt{n + \sqrt{n}} = \sqrt{k^2 + k}$$

Since $k \in \mathbb{N}$, we have the strict inequalities:

$$k^2 < k^2 + k < k^2 + 2k + 1 = (k + 1)^2$$

This places $k^2 + k$ strictly between two consecutive perfect squares, k^2 and $(k+1)^2$. Therefore, $k^2 + k$ cannot be a perfect square.

By part (a), $\sqrt{k^2 + k}$ must be irrational, which contradicts our assumption that $\sqrt{n + \sqrt{n}} = r \in \mathbb{Q}$.

Thus, $\sqrt{n + \sqrt{n}}$ is irrational for every $n \in \mathbb{N}$.